

Assignment 3

Title: Implement Min, Max, Sum and Average operations using Parallel Reduction.

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```
#include <iostream>
//#include <vector>
#include <omp.h>
#include <climits>
using namespace std;
void min_reduction(int arr[], int n) {
    int min_value = INT_MAX;
    #pragma omp parallel for reduction(min: min_value)
    for (int i = 0; i < n; i++) {
        if (arr[i] < min_value) {
            min_value = arr[i];
        }
    }
    cout << "Minimum value: " << min_value << endl;
}

void max_reduction(int arr[], int n) {
    int max_value = INT_MIN;
    #pragma omp parallel for reduction(max: max_value)
    for (int i = 0; i < n; i++) {
        if (arr[i] > max_value) {
            max_value = arr[i];
        }
    }
    cout << "Maximum value: " << max_value << endl;
}

void sum_reduction(int arr[], int n) {
    int sum = 0;
    #pragma omp parallel for reduction(+: sum)
    for (int i = 0; i < n; i++) {
        sum += arr[i];
    }
    cout << "Sum: " << sum << endl;
}

void average_reduction(int arr[], int n) {
```

```

int sum = 0;
#pragma omp parallel for reduction(+: sum)
for (int i = 0; i < n; i++) {
    sum += arr[i];
}
cout << "Average: " << (double)sum / (n-1) << endl;
}

int main() {
    int *arr,n;
    cout<<"\n enter total no of elements=>";
    cin>>n;
    arr=new int[n];
    cout<<"\n enter elements=>";
    for(int i=0;i<n;i++)
    {
        cin>>arr[i];
    }

    // int arr[] = {5, 2, 9, 1, 7, 6, 8, 3, 4};
    // int n = size(arr);

    min_reduction(arr, n);
    max_reduction(arr, n);
    sum_reduction(arr, n);
    average_reduction(arr, n);
}

```

Output:

```

datanalytics2@datanalytics2-OptiPlex-3050:~$ g++ -fopenmp Ass3.cpp -o output
datanalytics2@datanalytics2-OptiPlex-3050:~$ ./output
Enter total number of elements: 5
Enter elements:
10 20 30 40 50

Results using Parallel Reduction:
Minimum value: 10
Maximum value: 50
Sum: 150
Average: 30
datanalytics2@datanalytics2-OptiPlex-3050:~$ █

```